

FutureWorlds: LEARNING ROBOTIC WORLD MODELS FROM ALTERNATIVE FUTURES

Anonymous authors

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ABSTRACT

Robotic world models predict action-conditioned future scenes, providing a foundation for understanding action outcomes. However, turning alternative predictions into useful learning signals remains challenging: similar candidates limit informative quality comparisons, while diverging trajectories require persistent maintenance of their individual histories. We introduce FutureWorlds, a framework that unifies candidate construction, history maintenance, and learning from relative quality. Built on a multimodal discrete autoregressive model, FutureWorlds uses diverse beam search during reinforcement learning to construct candidate futures that balance confidence and diversity. Candidate-specific bounded memory preserves scene states and ensures that generation and policy scoring use matching histories. We further propose MemSP0 (Memory-Conditioned Search-Guided Policy Optimization), which converts video trajectory rewards into group-relative advantages to optimize the world model. On RT-1, BridgeV2, and RoboCasa, FutureWorlds reduces LPIPS for 32-frame predictions by 14.78%, 20.84%, and 9.12%, respectively, relative to the strongest baseline on each dataset. Under fixed evaluation configurations, only 200 MemSP0 updates further improve generation quality and support continued prediction beyond the training horizon. Memory ablations, decoding sensitivity analysis, and optical-flow evaluation show that these gains extend beyond visual quality to more accurate motion prediction and more consistent object states.

1 INTRODUCTION

Robotic world models predict how scenes evolve in response to actions, providing a foundation for action-outcome prediction (Finn et al., 2016), visual planning (Ebert et al., 2018; Hafner et al., 2019), behavior learning (Hafner et al., 2020), and policy evaluation (Yin et al., 2026; GigaWorld Team et al., 2026). Given the same observations, action sequence, and instruction, a model can propose multiple predictions of the future. These *alternative futures* represent different hypotheses about the scene’s evolution, with varying robot motions, object states, and prediction quality. Their differences offer a learning opportunity: by constructing useful alternatives and comparing their outcomes against recorded observations, a world model can learn from the predictions it generates.

Most autoregressive world models learn from recorded trajectories through supervised training. Their reliance on ground-truth histories during training differs from the self-generated histories available at inference, allowing errors to accumulate along a rollout (Ross et al., 2011; Bengio et al., 2015). Token-level likelihood also does not directly optimize complete generated sequences (Ranzato et al., 2016; Rennie et al., 2017). Figure 1 illustrates this challenge: supervised fine-tuning (SFT) can leave predictions with insufficient robot motion or inaccurate poses. Reinforcement learning (RL) connects generated outputs to sequence-level rewards (Black et al., 2024). In world modeling, RLVR-World uses decoded prediction metrics (Wu et al., 2025a), while RLIR introduces inverse-dynamics rewards for action following (Ye et al., 2025). These advances enable learning from trajectory feedback, but the resulting learning signal also depends on *which futures the model compares*. More candidates alone do not guarantee useful comparative feedback.

Learning from alternative futures requires coordinated candidate construction and history management. Candidate quality and diversity depend on the generation strategy (Holtzman et al., 2020): independent sampling does not explicitly coordinate these properties across candidates, while conventional beam search can concentrate on nearly identical sequences (Vijayakumar et al., 2018). A useful candidate set therefore needs to balance model confidence with distinct predictions. As these predictions unfold, each branch develops its own robot and object states. Retaining the corresponding history becomes especially important when prediction extends beyond the pretraining horizon, motivating bounded memory for sustained generation (Dai et al., 2019; Wu et al., 2025b; Zheng et al., 2026). Learning introduces a further requirement: policy scoring must condition on the same retained history that produces each candidate. These considerations connect three roles—*constructing* informative alternatives, *maintaining* their individual histories, and *learning* from their relative quality.



Figure 1: *Learning from alternative futures.* RT-1 comparison at +32; metrics use full images. FutureWorlds constructs diverse futures, maintains their histories, and learns through MemSPO by comparing trajectory rewards.



How can we construct informative alternative futures, maintain their individual histories, and learn from their relative quality to improve robotic world models?

We introduce FutureWorlds, a framework that learns robotic world models by constructing, maintaining, and comparing alternative futures (Figure 2). We instantiate the framework with a multimodal discrete autoregressive model (van den Oord et al., 2017; Wu et al., 2024), whose token probabilities support candidate expansion and trajectory scoring under visual, action, and language conditioning. *To construct alternatives*, diverse beam search combines model likelihood with cross-candidate differences during RL training. *To maintain their histories*, candidate-specific memory retains scene and state anchors alongside rolling context, supporting longer predictions within a bounded memory budget. Search and memory operate together as each candidate evolves, and generation and policy scoring use matching histories. *To compare and learn*, MemSPO (Memory-Conditioned Search-Guided Policy Optimization) converts video trajectory rewards into group-relative advantages and updates the model through a search-guided surrogate based on GRPO (Shao et al., 2024) and clipped policy optimization (Schulman et al., 2017). Candidate quality differences thus become feedback for improving the model that generates them.

Our evaluation covers RT-1 (Brohan et al., 2023), BridgeData V2 (Walke et al., 2023), and RoboCasa (Nasiriany et al., 2026). On 32-frame predictions, FutureWorlds reduces LPIPS by 14.78%, 20.84%, and 9.12% relative to the strongest baseline on each dataset. Under fixed evaluation configurations, only 200 MemSPO updates improve generation and support prediction beyond the training horizon. Memory ablations, decoding sensitivity analysis, and motion evaluation further assess historical context, robustness, and dynamic consistency. Our contributions are summarized as follows:

- 1 **Learning from Alternative Futures.** We introduce FutureWorlds, a framework that organizes model-generated futures into training experiences through coordinated candidate construction, history maintenance, and relative comparison.
- 2 **Search-Guided, History-Consistent Learning.** We propose MemSPO, a GRPO-based post-training algorithm that combines diverse candidate search with candidate-specific memory and matches the histories used for generation and policy scoring.

- ③ *Experimental Evaluation.* We evaluate generation quality and long-horizon prediction, analyze memory and decoding robustness, and assess motion consistency and inference resource requirements. LPIPS decreases by 9.12–20.84% versus the strongest baseline on each dataset.

2 RELATED WORK

Robotic World Models. Robotic world models predict action consequences for planning and policy learning. Early work uses action-conditioned video prediction for visual control (Finn et al., 2016; Ebert et al., 2018); PlaNet (Hafner et al., 2019) and Dreamer (Hafner et al., 2020) learn latent dynamics. UniSim (Yang et al., 2024) combines language and low-level controls for visual simulation, while Vid2World (Huang et al., 2026) adapts video diffusion for interactive prediction. VideoGPT (Yan et al., 2021) and iVideoGPT (Wu et al., 2024) develop discrete autoregressive models, with iVideoGPT integrating observations, actions, and rewards through compressive tokenization. FutureWorlds uses discrete autoregression to construct and compare alternative futures under shared visual, action, and language conditions.

Memory for Long-Horizon World Modeling. Transformer-XL (Dai et al., 2019) propagates segment-level history, while StreamingLLM (Xiao et al., 2024) retains initial attention sinks and recent context. StreamingT2V (Henschel et al., 2025) combines short- and long-term memory; FramePack (Zhang & Agrawala, 2025) packs historical frames into a fixed context budget. Wu et al. (2025b) introduce geometry-grounded memory for scene revisits. Ctrl-World (Guo et al., 2026) and Mem-World (Zheng et al., 2026) retrieve robotic observations through pose conditioning and spatiotemporal surfel indexing, respectively. FutureWorlds maintains bounded, candidate-specific histories for long-horizon prediction and uses matching contexts for generation and policy scoring, preserving distinct conditioning as candidate futures diverge.

Search-Guided Reinforcement Learning for World Models. MIXER (Ranzato et al., 2016) and SCST (Rennie et al., 2017) optimize sequence metrics through policy gradients; DDPO (Black et al., 2024) extends reward optimization to diffusion models. RLVR-World (Wu et al., 2025a) rewards decoded prediction quality, while RLIR (Ye et al., 2025) evaluates action following through inverse dynamics. Candidate construction determines which predictions provide this feedback. Diverse Beam Search (Vijayakumar et al., 2018) reduces redundancy, Expert Iteration (Anthony et al., 2017) alternates search and policy learning, and BGRPO (Lee et al., 2025) combines beam-generated candidates with rank-aware rewards for algebraic tasks. QWM (Dong et al., 2026) searches predicted futures for robotic action selection and learns policies and values from real transitions. FutureWorlds instead searches futures under a fixed action sequence. MemSP0 updates the video predictor from their relative quality using history-consistent scoring and a search-guided surrogate based on GRPO’s group-relative advantages and clipped updates (Shao et al., 2024).

3 METHOD

Figure 2 presents FutureWorlds, which learns from alternative futures through coordinated candidate construction, history maintenance, and relative comparison. A multimodal autoregressive world model first learns conditional video prediction through supervised training. During post-training, diverse search constructs candidate futures under shared visual, action, and language conditions. Each candidate retains its own history, with bounded persistent memory supporting continuation beyond the pretraining horizon. MemSP0 compares decoded trajectories through video rewards and converts their relative quality into world-model updates. Generation and policy scoring use matching candidate histories, while true future frames enter only reward computation during post-training.

3.1 PROBLEM FORMULATION

Robotic world modeling predicts how a scene evolves under actions. We use a GPT-style autoregressive model to represent alternative future trajectories conditioned on the same visual observations, action sequence, and optional language instruction. Let $\mathbf{z}_{1:H}$ denote the discrete visual sequence for H future frames, with N dynamic tokens per frame. Given initial visual conditions $\mathbf{c}_v$, robot actions $\mathbf{a}_{0:H-1}$, and an optional language instruction ℓ , the world model factorizes the conditional

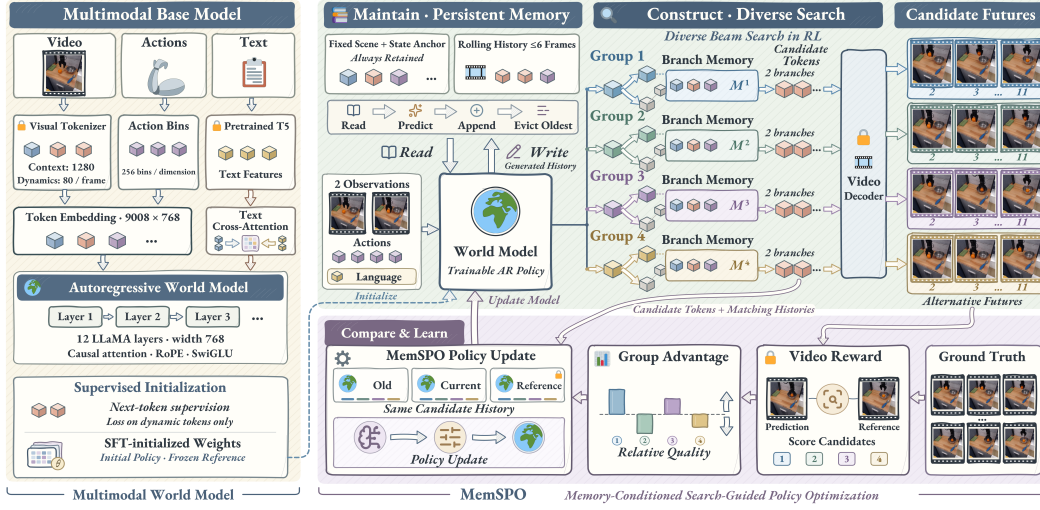


Figure 2: *Overview of FutureWorlds.* Supervised initialization trains the multimodal world model. MemSPO searches alternative futures, maintains separate histories, and learns from relative trajectory quality. Generation and policy scoring use matching histories.

distribution as

$$\pi_{\theta}(\mathbf{z}_{1:H} \mid \mathbf{c}_v, \mathbf{a}_{0:H-1}, \ell) = \prod_{t=1}^H \prod_{j=1}^N \pi_{\theta}(z_{t,j} \mid \mathbf{c}_v, \mathbf{z}_{<t}, \mathbf{z}_{t,<j}, \mathbf{a}_{<t}, \ell). \quad (1)$$

Here, $\mathbf{z}_{<t}$ contains tokens from preceding frames, $\mathbf{z}_{t,<j}$ is the within-frame prefix, and $\mathbf{a}_{<t} = \mathbf{a}_{0:t-1}$ contains actions available before frame t . In particular, a_{t-1} conditions the transition to that frame. A visual decoder maps predicted tokens back to video. This factorization supports both candidate expansion and trajectory scoring. Our goal is to construct alternative predictions under the same conditions, maintain their histories within a bounded context, and learn from their relative quality.

3.2 MULTIMODAL AUTOREGRESSIVE WORLD MODEL

The multimodal model provides the representation for constructing and scoring candidate futures. Following iVideoGPT’s conditional video compression (Wu et al., 2024), we use more context tokens for the initial scene and fewer dynamic tokens for subsequent changes, reducing repeated encoding of static information. Context and conditional dynamics encoders produce discrete representations through Finite Scalar Quantization (FSQ; Mentzer et al., 2024), and a visual decoder reconstructs frames:

$$C = Q_c(E_{\phi}^c(o_{\text{ref}})), \quad \mathbf{z}_t = Q_d(E_{\phi}^d(o_t \mid o_{\text{ref}})), \quad \tilde{o}_t = D_{\phi}(C, \mathbf{z}_t). \quad (2)$$

Here, o_{ref} is the initial reference frame, ϕ denotes tokenizer parameters, and Q_c and Q_d map continuous features to discrete tokens with straight-through gradients.

The tokenizer jointly optimizes pixel, perceptual, and adversarial losses:

$$\mathcal{L}_{\text{tok}}(\phi) = \lambda_{\text{rec}} \mathcal{L}_1 + \lambda_{\text{perc}} \mathcal{L}_{\text{LPIPS}} + \lambda_{\text{adv}} \mathcal{L}_{\text{GAN}}. \quad (3)$$

Pixel and perceptual losses supervise both reference and dynamic frames; the adversarial term further constrains visual quality. The tokenizer remains frozen throughout world-model SFT and RL. Dataset-specific initialization and adaptation details are given in Appendix B.1.

As Figure 2 (left) shows, actions undergo per-dimension normalization and discretization, then form a temporally ordered causal sequence with visual tokens. A frozen T5 encoder (Raffel et al., 2020) supplies language features through cross-attention. Supervised fine-tuning conditions on ground-truth histories and applies negative log-likelihood only to target dynamic tokens:

$$\mathcal{L}_{\text{SFT}}(\theta) = -\mathbb{E}_{\mathcal{D}} \left[\frac{1}{HN} \sum_{t=1}^H \sum_{j=1}^N \log \pi_{\theta}(z_{t,j} \mid \mathbf{c}_v, \mathbf{z}_{<t}, \mathbf{z}_{t,<j}, \mathbf{a}_{<t}, \ell) \right]. \quad (4)$$

This objective jointly updates the autoregressive backbone, action embeddings, and text cross-attention modules. The resulting model initializes the post-training policy, with a frozen copy serving as the reference policy.

3.3 CANDIDATE-SPECIFIC PERSISTENT MEMORY

As candidate futures diverge, each accumulates its own prediction history. Persistent memory supports long-horizon continuation within a bounded context by retaining the initial scene, a state anchor, and recent predictions for each candidate. Let $\mathbf{c}_v = (C, \mathbf{z}_{\text{anc}})$ and let $\tilde{\mathbf{a}}_s$ denote discrete action tokens. Before predicting frame t , candidate i has memory

$$M_t^{(i)} = \left[C, \mathbf{z}_{\text{anc}}, \tilde{\mathbf{a}}_0, \left\{ (\hat{\mathbf{z}}_s^{(i)}, \tilde{\mathbf{a}}_s) \right\}_{s \in \mathcal{I}_t} \right], \quad \mathcal{I}_t = \{\max(1, t - W), \dots, t - 1\}. \quad (5)$$

The rolling history is temporally ordered and initially empty; its window length W satisfies the token budget $|M_t^{(i)}| \leq B$. Scene and state anchors persist throughout prediction. After each frame, memory appends its dynamic tokens and corresponding action, excluding older history from the model context once the window is full. Language remains available independently of this history.

During search, branches inherit their parents’ memories, which are reordered together with candidate selection. To learn from each resulting future under the conditions that generate it, policy scoring uses the same memory and within-frame prefix:

$$\log \pi_\theta(\hat{\mathbf{z}}_{1:H}^{(i)} \mid \mathbf{c}_v, \mathbf{a}_{0:H-1}, \ell) = \sum_{t=1}^H \sum_{j=1}^N \log \pi_\theta(\hat{z}_{t,j}^{(i)} \mid M_t^{(i)}, \hat{\mathbf{z}}_{t,<j}^{(i)}, \ell). \quad (6)$$

The old, current, and reference policies share these conditions, preserving the link between each candidate and its history during optimization. Supervised training uses ground-truth histories; RL and free rollouts use predicted histories, with true future frames reserved for reward computation.

3.4 MEMORY-CONDITIONED SEARCH-GUIDED POLICY OPTIMIZATION

MemSP0 turns alternative futures into post-training feedback: it constructs candidates through diverse search, maintains their histories as described above, and learns from their relative trajectory quality. The update uses GRPO-style group-relative advantages and a clipped policy surrogate (Shao et al., 2024; Schulman et al., 2017).

Constructing alternative futures. We maintain G search groups with b beams each. For prefix y and next token v , group g uses the diverse search score (Vijayakumar et al., 2018):

$$S_g(y \oplus v) = S_g(y) + \log \pi_{\text{old}}(v \mid u(y)) - \lambda_{\text{div}} d_g(v). \quad (7)$$

Here, $u(y)$ contains the branch’s memory and multimodal conditions, and $d_g(v)$ measures the normalized frequency of v selected by earlier groups at the same position. Each group retains its b highest-scoring branches and returns one final trajectory, producing G videos under the same input conditions. The penalty encourages token-level differences across groups and affects selection only; policy scoring uses unmodified probabilities.

Comparing trajectory quality. The frozen visual decoder converts candidate tokens into videos. Following RLVR-World (Wu et al., 2025a), we combine pixel error and perceptual distance (Zhang et al., 2018) to score each trajectory:

$$R_i = -\frac{1}{H} \sum_{t=1}^H \left[\text{MAE}(\hat{o}_t^{(i)}, o_t) + \text{LPIPS}(\hat{o}_t^{(i)}, o_t) \right]. \quad (8)$$

We normalize rewards within each group of candidate futures for the same input:

$$A_i = \frac{R_i - \bar{R}}{\text{Std}(R_1, \dots, R_G) + \varepsilon}, \quad \bar{R} = \frac{1}{G} \sum_{i=1}^G R_i. \quad (9)$$

All dynamic tokens in a trajectory share A_i . True future frames enter reward computation only, never candidate memory. Video decoding and reward computation carry no gradients.

Learning under matching histories. The current, old, and frozen reference policies share each candidate’s generation condition $u_{t,j}^{(i)} = (M_t^{(i)}, \hat{z}_{t,<j}^{(i)}, \ell)$. The token probability ratio is

$$\rho_{t,j}^{(i)}(\theta) = \frac{\pi_{\theta}(\hat{z}_{t,j}^{(i)} | u_{t,j}^{(i)})}{\pi_{\text{old}}(\hat{z}_{t,j}^{(i)} | u_{t,j}^{(i)})}. \quad (10)$$

MemSPO maximizes the surrogate objective

$$\mathcal{J}_{\text{MemSPO}}(\theta) = \frac{1}{GHN} \sum_{i=1}^G \sum_{t=1}^H \sum_{j=1}^N \left[\ell_{\text{clip}}(\rho_{t,j}^{(i)}(\theta), A_i) - \beta \kappa_{t,j}^{(i)}(\theta) \right]. \quad (11)$$

Here, ℓ_{clip} is the clipped policy term, including dual clipping for negative advantages. The reference regularizer is $\kappa = q - \log q - 1$, where $q = \pi_{\text{ref}}/\pi_{\theta}$ evaluates the same token under the same history.

Each iteration constructs fresh candidates and performs one world-model update, transferring their relative-quality feedback into the model parameters. The visual tokenizer, text encoder, and memory rules remain fixed. Beam selection changes the candidate distribution, so this is a search-guided surrogate without an unbiased on-policy gradient guarantee. We analyze how candidate construction shapes the group-relative reward gradient and bound its magnitude under candidate collapse (Appendix A, Proposition A.1 and Corollary A.1).

4 EXPERIMENTS

Our experiments address four research questions: **❶ RQ1** How well does FutureWorlds predict future observations? **❷ RQ2** Can MemSPO improve generation with few post-training updates? **❸ RQ3** Do generation gains translate into more accurate motion and dynamics? **❹ RQ4** Does memory maintenance improve prediction beyond the training horizon?

4.1 EXPERIMENTAL SETUP

Baselines. We compare four baseline categories: ♦ video generation models: FitVid (Babaeizadeh et al., 2021) and iVideoGPT (Wu et al., 2024); ♦ action-conditioned world models: IRASim (Zhu et al., 2025), Vid2World (Huang et al., 2026), Ctrl-World (Guo et al., 2026), DriftWorld (Lu et al., 2026), and WEAVER (Jain et al., 2026); ♦ RL post-training methods: RLVR-World (Wu et al., 2025a) and PersistWorld (Bardhan et al., 2026); and ♦ adapted video foundation models: Wan2.2-T12V-5B (Team Wan et al., 2025), Cosmos-Predict2.5-2B (NVIDIA, 2025), and DreamDojo-2B (Gao et al., 2026). Adaptation and evaluation details appear in Appendix B.3.

Datasets and Evaluation. We evaluate action-conditioned video generation on RT-1 (Brohan et al., 2023), BridgeV2 (Walke et al., 2023), and RoboCasa (Nasiriany et al., 2026), covering real and simulated manipulation. Table 1 compares 32-frame predictions on 128 fixed held-out trajectories per dataset using PSNR, SSIM, and LPIPS (Zhang et al., 2018). Methods share trajectories and target frames, with predictions resized to the reference resolution. Splits, preprocessing, and model-specific inputs appear in the appendix.

Implementation Details. We train FutureWorlds in PyTorch on 16 NVIDIA H20 GPUs across two servers. Starting from SFT, MemSPO uses AdamW with learning rate 10^{-6} and freezes the visual tokenizer and text encoder. Diverse beam search uses four groups of two beams, retaining one trajectory per group for reward comparison and updates. Main evaluations use a beam width of four and persistent memory. SFT settings, dataset-specific budgets, and remaining hyperparameters appear in the appendix.

4.2 VIDEO GENERATION QUALITY (RQ1)

In Table 1, FutureWorlds leads all three metrics on all datasets. Relative to the strongest baseline per metric on BridgeV2, it gains 1.71 dB PSNR and 2.17% SSIM and reduces LPIPS by 20.84%. LPIPS also decreases by 14.78% on RT-1 and 9.12% on RoboCasa relative to the strongest baseline per dataset. Figure 3 shows closer object locations and arm poses in RT-1, fewer object substitutions and local distortions in BridgeV2, and more accurate arm and gripper poses in RoboCasa.

Table 1: Quantitative comparison of 32-frame action-conditioned video prediction (128 trajectories per dataset). Metrics: PSNR $\uparrow$, SSIM $\uparrow$, and LPIPS $\downarrow$. 1st, 2nd, and 3rd mark the top three scores; the last row reports relative gains over the runner-up.

Method	RT-1			BridgeV2			RoboCasa		
	PSNR	SSIM	LPIPS	PSNR	SSIM	LPIPS	PSNR	SSIM	LPIPS
<i>Video Generation Models</i>									
FitVid	18.71	0.5916	0.4750	17.09	0.6072	0.5235	13.50	0.5422	0.5655
iVideoGPT	15.96	0.5571	0.3833	20.06	0.7791	0.1899	16.67	0.7187	0.2366
<i>Action-Conditioned World Models</i>									
IRASim	16.83	0.5936	0.3474	17.39	0.7094	0.2524	13.55	0.5834	0.3706
Vid2World	19.03	0.7210	0.1960	18.92	0.7110	0.1880	15.40	0.7000	0.2287
Ctrl-World	19.11	0.7110	0.2522	18.38	0.6932	0.2621	15.55	0.7035	0.2686
DriftWorld	20.54	0.7410	0.2595	20.32	0.7429	0.2379	16.50	0.7200	0.2891
WEAVER	19.74	0.7617	0.2321	18.74	0.7447	0.2476	14.62	0.7136	0.2726
<i>RL Post-Training</i>									
RLVR-World	19.47	0.7360	0.1820	19.06	0.7210	0.1780	15.80	0.7100	0.2017
PersistWorld	18.77	0.6995	0.2717	19.07	0.7154	0.2408	15.61	0.7112	0.2639
<i>Adapted Video Foundation Models</i>									
Wan2.2-TI2V-5B	18.42	0.7058	0.2478	16.35	0.6889	0.2568	14.37	0.6906	0.2711
Cosmos-Predict2.5-2B	17.77	0.6765	0.2889	18.98	0.7799	0.1870	14.30	0.6802	0.2916
DreamDojo-2B	18.47	0.7152	0.2660	16.95	0.7461	0.2563	13.92	0.6349	0.3278
FutureWorlds (ours)	22.47	0.8026	0.1551	22.03	0.7968	0.1409	18.47	0.7659	0.1833
<i>Gain over runner-up</i>	+9.40%	+5.37%	+14.78%	+8.42%	+2.17%	+20.84%	+10.80%	+6.38%	+9.12%

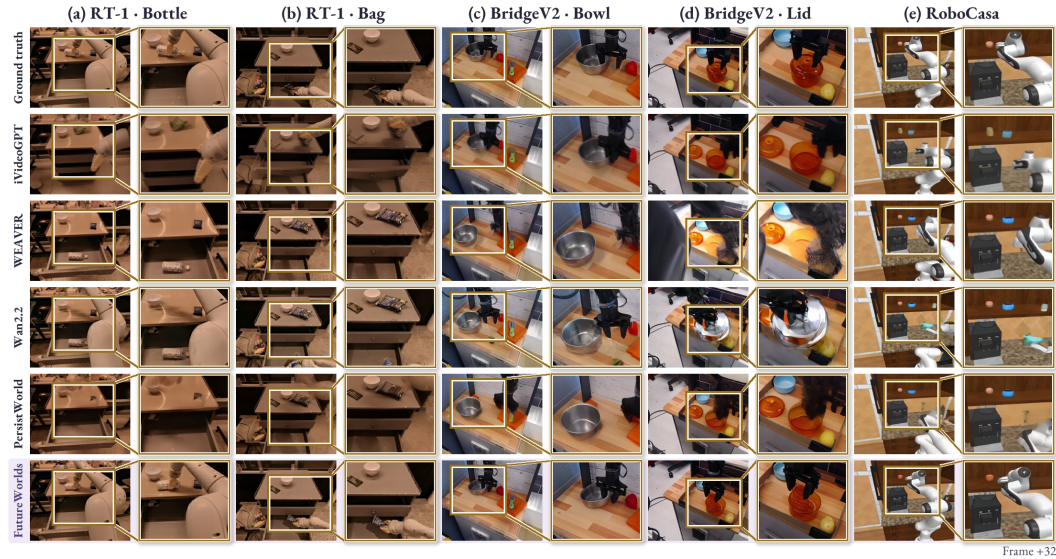


Figure 3: **Qualitative comparison at frame 32.** Five selected cases on RT-1, BridgeV2, and RoboCasa comparing four baselines and our method, with full images and matched detail views.

These comparisons support improved object consistency and action-conditioned prediction. The matched crops highlight errors that scene appearance alone can obscure: an object can retain a plausible texture while occupying the wrong position relative to the gripper. Preserving this relation helps distinguish faithful action-conditioned prediction from visually plausible but inconsistent manipulation states. Further quantitative and qualitative results appear in the appendix.

Table 2: Post-training comparison with 200 updates. Metrics: PSNR $\uparrow$, SSIM $\uparrow$, and LPIPS $\downarrow$. Gain rows show relative improvements over GRPO (%).

Method	10 frames			20 frames			32 frames		
	PSNR	SSIM	LPIPS	PSNR	SSIM	LPIPS	PSNR	SSIM	LPIPS
RT-1									
SFT (50k)	23.76	0.8298	0.1308	22.49	0.8075	0.1527	21.37	0.7860	0.1734
GRPO (200)	23.97	0.8319	0.1289	22.60	0.8089	0.1517	21.39	0.7858	0.1739
MemSPO (200)	24.33	0.8357	0.1240	23.42	0.8192	0.1397	22.45	0.8016	0.1565
Gain over GRPO	+1.49%	+0.45%	+3.80%	+3.61%	+1.28%	+7.92%	+4.97%	+2.01%	+9.98%
BridgeV2									
SFT (50k)	24.07	0.8300	0.1096	22.74	0.8084	0.1270	21.73	0.7940	0.1450
GRPO (200)	24.12	0.8301	0.1089	22.80	0.8088	0.1263	21.80	0.7945	0.1441
MemSPO (200)	24.31	0.8320	0.1073	23.00	0.8108	0.1244	21.97	0.7962	0.1418
Gain over GRPO	+0.83%	+0.23%	+1.44%	+0.88%	+0.25%	+1.53%	+0.78%	+0.22%	+1.58%
RoboCasa									
SFT (50k)	19.50	0.7957	0.1513	18.22	0.7689	0.1787	17.50	0.7519	0.1971
GRPO (200)	19.68	0.7965	0.1514	18.64	0.7729	0.1758	18.03	0.7581	0.1919
MemSPO (200)	19.92	0.7984	0.1476	18.88	0.7743	0.1718	18.29	0.7600	0.1877
Gain over GRPO	+1.21%	+0.24%	+2.47%	+1.31%	+0.19%	+2.28%	+1.45%	+0.25%	+2.17%

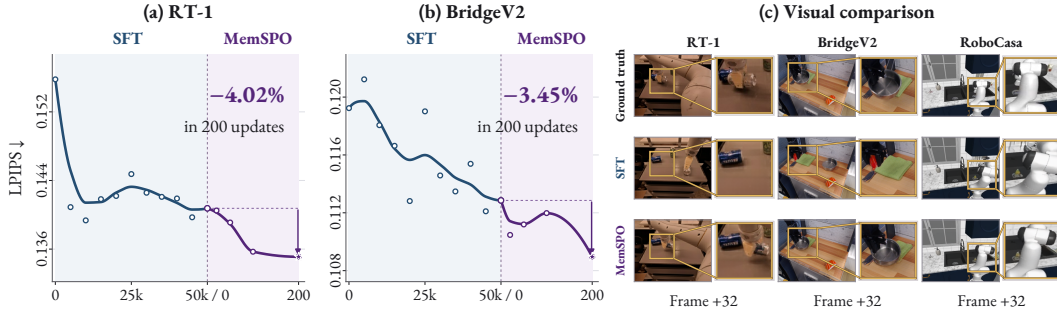


Figure 4: **Post-training gains and qualitative improvements.** (a,b) LPIPS during SFT and MemSPO training. (c) Frame-32 predictions with matched detail views on three datasets.

4.3 POST-TRAINING EFFECTIVENESS AND EFFICIENCY (RQ2)

On 128 matched trajectories per dataset with decoding and memory fixed, MemSPO leads all three metrics at 10, 20, and 32 frames after 200 RL updates (Table 2). At 32 frames, LPIPS drops by 9.98%, 1.58%, and 2.17% versus GRPO on RT-1, BridgeV2, and RoboCasa, showing gains beyond ten-frame training.

On 32 monitoring clips, 200 updates reduce LPIPS by 4.02%/3.45% on RT-1/BridgeV2 versus SFT (Figure 4a,b), with improved object consistency and motion (c). Gains persist across five decoding settings (Appendix Table 3).

With checkpoints and beam width fixed, the LPIPS reduction remains 3.03–4.32% on RT-1 and 2.60–3.84% on BridgeV2 across the tested temperature and top- p settings. This consistency indicates that MemSPO’s gains are not confined to the default decoding configuration, complementing the matched comparison with GRPO under matched inputs, memory settings, and rollout lengths.

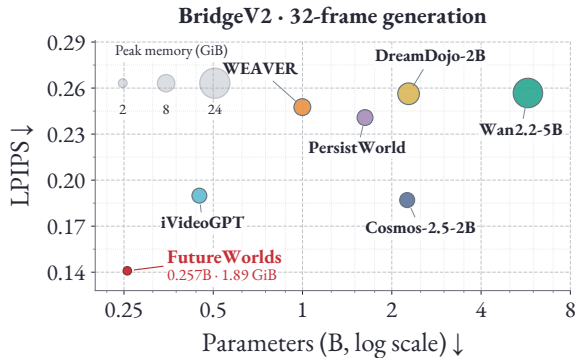


Figure 5: **Model size and quality.** Bubble area denotes peak GPU memory on one H20.

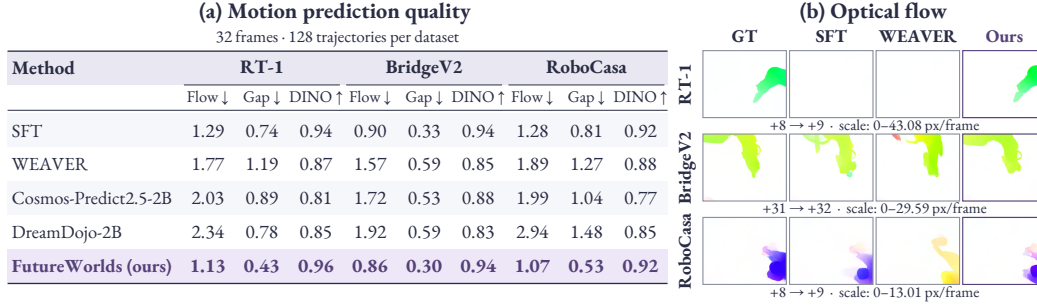


Figure 6: **Motion quality and optical-flow comparisons.** (a) Full-cohort results at 32 frames. (b) Selected examples with a shared flow-color scale within each dataset.

Resource footprint. FutureWorlds leads seven models in BridgeV2 LPIPS with 0.257B inference parameters (visual codec included) and 1.89 GiB peak memory (Figure 5). Costs use three 32-frame rollouts at batch size one, with native configurations and cached language features. Beam decoding takes 26.76 s versus iVideoGPT’s 4.14 s per rollout. Thus, smaller parameter and memory footprints do not ensure faster inference, motivating decoding acceleration.

4.4 MOTION AND DYNAMIC PREDICTION (RQ3)

Using WorldArena’s flow and feature components (Shang et al., 2026), we evaluate 128 trajectories per dataset. FutureWorlds leads all three diagnostics among five methods (Figure 6a). Flow error decreases by 12.40%, 4.57%, and 16.68% versus SFT on RT-1, BridgeV2, and RoboCasa. The examples show motion closer to recorded futures (b), supporting gains beyond pixel fidelity.

4.5 MEMORY MAINTENANCE (RQ4)

With weights and decoding fixed, full memory leads PSNR, SSIM, and MSE at 32 frames across all datasets (Figure 7). It reduces MSE by 40.83%, 57.69%, and 29.22% versus removing historical memory on RT-1, BridgeV2, and RoboCasa. Recent history supplies most gains; the initial anchor adds further benefits across real and simulated manipulation.

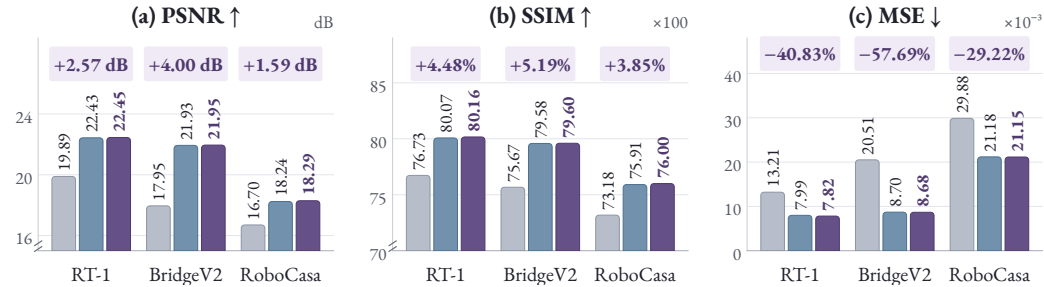


Figure 7: **Memory ablation at 32 frames.** ■ w/o Historical Memory, ■ w/o Initial Anchor, and ■ Full Model. Highlighted gains compare the full model with historical memory removed. PSNR and SSIM axes are truncated as marked.

5 CONCLUSION

We present FutureWorlds, which learns from alternative futures through coordinated candidate construction, history maintenance, and quality comparison. MemSP0 combines diverse search with history-consistent optimization, improving generation within hundreds of updates and beyond the training horizon. Three-dataset evaluations support gains in visual quality, object consistency, and motion prediction. BridgeV2 comparisons also show better quality with fewer parameters and lower measured memory usage, although autoregressive beam search incurs latency. Future work focuses on faster decoding and validating these benefits in closed-loop planning and robot control.

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A THEORETICAL ANALYSIS

We analyze how alternative futures supply a learning signal in FutureWorlds. On a fixed candidate set, the local reward gradient of MemSP0 admits a pairwise decomposition that links relative rewards to differences in policy score gradients. The analysis characterizes this update; it does not establish that search produces better candidates.

A.1 PAIRWISE STRUCTURE OF THE REWARD GRADIENT

For a fixed input, consider $G \geq 2$ candidates with $L = HN$ tokens each. Flatten the token indices into $k \in \{1, \dots, L\}$, with $z_k^{(i)}$ and $u_k^{(i)}$ denoting candidate i 's token and condition, including its memory and within-frame prefix. Tokens, conditions, rewards, and normalization statistics remain fixed during differentiation. Selected tokens have positive old-policy probabilities, and current-policy probabilities are differentiable near θ_{old} .

Let $\sigma_R = \text{Std}(R_1, \dots, R_G)$ use the same convention as Eq. (9). Define the group-relative advantage and mean token score gradient as

$$A_i = \frac{R_i - \bar{R}}{\sigma_R + \varepsilon}, \quad \bar{R} = \frac{1}{G} \sum_{i=1}^G R_i, \quad \mathbf{s}_i = \frac{1}{L} \sum_{k=1}^L \nabla_{\theta} \log \pi_{\theta}(z_k^{(i)} | u_k^{(i)}) \Big|_{\theta=\theta_{\text{old}}}. \quad (\text{A.1})$$

Here, $\varepsilon > 0$ stabilizes reward normalization. The implementation uses sample standard deviation, so $\sigma_R^2 = \frac{1}{G-1} \sum_i (R_i - \bar{R})^2$. We first analyze the normalized advantages above; the numerical zero-variance guard is addressed below. The reward component of Eq. (11) is

$$\mathcal{J}_R(\theta) = \frac{1}{GL} \sum_{i=1}^G \sum_{k=1}^L \ell_{\text{clip}}(\rho_k^{(i)}(\theta), A_i), \quad \rho_k^{(i)}(\theta) = \frac{\pi_{\theta}(z_k^{(i)} | u_k^{(i)})}{\pi_{\text{old}}(z_k^{(i)} | u_k^{(i)})}. \quad (\text{A.2})$$

Proposition A.1 (Pairwise reward-gradient identity). *Suppose generation and policy scoring use matching candidate histories, the clipping radius satisfies $0 < \epsilon_c < 1$, and the negative-advantage dual-clipping constant satisfies $c_d > 1$. At $\theta = \theta_{\text{old}}$, the reward gradient admits the pairwise representation*

$$\mathbf{g}_R := \nabla_{\theta} \mathcal{J}_R(\theta) \Big|_{\theta=\theta_{\text{old}}} = \frac{1}{G} \sum_{i=1}^G A_i \mathbf{s}_i = \frac{\sum_{i < j} (R_i - R_j)(\mathbf{s}_i - \mathbf{s}_j)}{G^2(\sigma_R + \varepsilon)}. \quad (\text{A.3})$$

Proof. We first reduce the clipped objective locally. Write

$$v(\rho, A) = \min\{\rho A, \text{clip}(\rho, 1 - \epsilon_c, 1 + \epsilon_c)A\}. \quad (\text{A.4})$$

The policy term is $v(\rho, A)$ for $A \geq 0$ and $\max\{v(\rho, A), c_d A\}$ for $A < 0$. At identical parameters, evaluating numerator and denominator under the same fixed condition gives $\rho_k^{(i)}(\theta_{\text{old}}) = 1$. Matching this condition to generation additionally ensures that the score refers to the history that produces the candidate. In a neighborhood of this point, ordinary clipping is inactive. For $A_i < 0$, the strict inequality $A_i > c_d A_i$ also makes dual clipping inactive; for $A_i = 0$, the term and its derivative are zero. Consequently,

$$\nabla_{\theta} \ell_{\text{clip}}(\rho_k^{(i)}(\theta), A_i) \Big|_{\theta=\theta_{\text{old}}} = A_i \nabla_{\theta} \rho_k^{(i)}(\theta) \Big|_{\theta=\theta_{\text{old}}}. \quad (\text{A.5})$$

The denominator of the probability ratio is fixed, so its derivative satisfies

$$\nabla_{\theta} \rho_k^{(i)}(\theta) = \frac{\nabla_{\theta} \pi_{\theta}(z_k^{(i)} | u_k^{(i)})}{\pi_{\text{old}}(z_k^{(i)} | u_k^{(i)})} = \rho_k^{(i)}(\theta) \nabla_{\theta} \log \pi_{\theta}(z_k^{(i)} | u_k^{(i)}). \quad (\text{A.6})$$

Substituting into Eq. (A.2) and evaluating at the old policy yields

$$\mathbf{g}_R = \frac{1}{G} \sum_{i=1}^G A_i \mathbf{s}_i = \frac{1}{G(\sigma_R + \varepsilon)} \sum_{i=1}^G (R_i - \bar{R}) \mathbf{s}_i. \quad (\text{A.7})$$

To obtain the pairwise form, symmetry and the vanishing diagonal terms give

$$\sum_{i < j} (R_i - R_j)(\mathbf{s}_i - \mathbf{s}_j) = \frac{1}{2} \sum_{i=1}^G \sum_{j=1}^G (R_i \mathbf{s}_i + R_j \mathbf{s}_j - R_i \mathbf{s}_j - R_j \mathbf{s}_i). \quad (\text{A.8})$$

Collecting the four sums and using $\sum_i R_i = G\bar{R}$ gives

$$\sum_{i < j} (R_i - R_j)(\mathbf{s}_i - \mathbf{s}_j) = G \sum_{i=1}^G R_i \mathbf{s}_i - \left(\sum_{i=1}^G R_i \right) \left(\sum_{i=1}^G \mathbf{s}_i \right) = G \sum_{i=1}^G (R_i - \bar{R}) \mathbf{s}_i. \quad (\text{A.9})$$

Combining Eqs. (A.7) and (A.9) proves Eq. (A.3). $\square$

Each pair contributes through its reward and score-gradient differences; either vanishing difference removes its contribution. No independence assumption is required, so the identity also holds for beam-search candidates. The algebra applies to any fixed conditions shared by numerator and denominator; history matching gives these scores their intended candidate-generation interpretation, rather than providing an independent performance guarantee.

A.2 A BOUND UNDER CANDIDATE COLLAPSE

Define the empirical reward variance and score-gradient dispersion by

$$\bar{\mathbf{s}} = \frac{1}{G} \sum_{i=1}^G \mathbf{s}_i, \quad V_R = \frac{1}{G} \sum_{i=1}^G (R_i - \bar{R})^2, \quad V_s = \frac{1}{G} \sum_{i=1}^G \|\mathbf{s}_i - \bar{\mathbf{s}}\|_2^2. \quad (\text{A.10})$$

We distinguish V_R from σ_R^2 so that the result accommodates either population or sample standard deviation in advantage normalization.

Corollary A.1 (Reward-gradient magnitude). *Under the conditions of Proposition A.1,*

$$\|\mathbf{g}_R\|_2 \leq \frac{\sqrt{V_R V_s}}{\sigma_R + \varepsilon}. \quad (\text{A.11})$$

In particular, the reward gradient is zero if all candidate rewards are equal or all candidate score gradients coincide.

Proof. Since $\sum_i (R_i - \bar{R}) = 0$, subtracting the mean score gradient leaves Eq. (A.7) unchanged:

$$\mathbf{g}_R = \frac{1}{G(\sigma_R + \varepsilon)} \sum_{i=1}^G (R_i - \bar{R})(\mathbf{s}_i - \bar{\mathbf{s}}). \quad (\text{A.12})$$

The triangle inequality gives

$$\|\mathbf{g}_R\|_2 \leq \frac{1}{G(\sigma_R + \varepsilon)} \sum_{i=1}^G |R_i - \bar{R}| \|\mathbf{s}_i - \bar{\mathbf{s}}\|_2. \quad (\text{A.13})$$

Applying the Cauchy–Schwarz inequality to the two scalar sequences yields

$$\|\mathbf{g}_R\|_2 \leq \frac{\sqrt{\sum_i (R_i - \bar{R})^2} \sqrt{\sum_i \|\mathbf{s}_i - \bar{\mathbf{s}}\|_2^2}}{G(\sigma_R + \varepsilon)} = \frac{\sqrt{V_R V_s}}{\sigma_R + \varepsilon}. \quad (\text{A.14})$$

If either V_R or V_s vanishes, the upper bound is zero. $\square$

We can also express dispersion through pairwise distances. Expanding the squared norms gives

$$\sum_{i < j} \|\mathbf{s}_i - \mathbf{s}_j\|_2^2 = G \sum_{i=1}^G \|\mathbf{s}_i\|_2^2 - \left\| \sum_{i=1}^G \mathbf{s}_i \right\|_2^2 = G \sum_{i=1}^G \|\mathbf{s}_i - \bar{\mathbf{s}}\|_2^2. \quad (\text{A.15})$$

Hence $V_s = G^{-2} \sum_{i < j} \|\mathbf{s}_i - \mathbf{s}_j\|_2^2$. If every pair satisfies $\|\mathbf{s}_i - \mathbf{s}_j\|_2 \leq \delta$, the $G(G-1)/2$ pairs imply

$$V_s \leq \frac{G-1}{2G} \delta^2, \quad \|\mathbf{g}_R\|_2 \leq \frac{\sqrt{V_R}}{\sigma_R + \varepsilon} \sqrt{\frac{G-1}{2G}} \delta. \quad (\text{A.16})$$

Thus, at a fixed reward scale, collapse in candidate score gradients limits the reward-driven update. Larger dispersion alone does not guarantee a larger or better-directed update, since pairwise contributions can cancel. Visual diversity is also distinct from score-gradient dispersion; connecting the two requires empirical analysis.

Scope. These statements concern the reward component at θ_{old} ; the reference-regularization gradient can remain nonzero under collapse, and clipping can become active away from this point. Because search changes the candidate distribution, the analysis implies neither an unbiased on-policy gradient nor monotonic improvement. The benefit of diverse beam search remains an empirical question. In our implementation, each fresh candidate set receives one update evaluated at the old parameters, where clipping is locally inactive. If $\sigma_R \leq 10^{-6}$, the implementation sets all advantages to zero; the reward gradient is then zero and the magnitude bound remains valid, although the unguarded pairwise equality need not hold for small nonzero reward differences. The full optimizer step also includes reference regularization and optimizer preconditioning, so it is distinct from $\mathbf{g}_R$.

B IMPLEMENTATION DETAILS

B.1 TOKENIZER INITIALIZATION AND ADAPTATION

We initialize the visual tokenizer from the released RT-1-pretrained checkpoint. For RT-1, we reuse this checkpoint directly, without an additional tokenizer adaptation stage. For Bridge-v2, we use a tokenizer adapted to Bridge-v2 from the same RT-1-pretrained initialization. In both datasets, the tokenizer is fixed before world-model SFT and remains frozen throughout SFT and MemSP0 post-training. These dataset-specific choices concern the visual representation used to initialize the world model; they do not change the MemSP0 update.

B.2 TRAINING ALGORITHM

Algorithm 1 summarizes one-input updates; distributed training averages the losses across inputs. The colors identify candidate construction, history maintenance, and relative-quality learning. The tokenizer, text encoder, and SFT reference policy remain frozen throughout post-training.

Algorithm 1 FutureWorlds: learning from alternative futures

Input: SFT parameters θ_0 ; training examples $x = (\mathbf{c}_v, \mathbf{a}, \ell, o_{1:H})$; horizon H ; N tokens/frame; G groups, b branches/group; memory budget B .

Output: Updated world-model parameters θ .

```

1 Initialize  $\theta \leftarrow \theta_0$  and freeze  $\pi_{\text{ref}} \leftarrow \pi_{\theta_0}$ .
2 For each post-training iteration do
3   Draw  $x$ ; set  $\theta_{\text{old}} \leftarrow \text{stopgrad}(\theta)$ .
4   Initialize  $G$  groups of  $b$  branches with the same scene and state anchors.
5   Set the first branch score in each group to 0 and the others to  $-\infty$ .
6   Construct / Maintain for  $t = 1, \dots, H$  do
7     Build each branch's prefix from its memory and actions; reset its KV cache.
8     For token position  $j = 1, \dots, N$  do
9       For group  $g = 1, \dots, G$  do
10        Compute valid-visual-token log probabilities with  $\pi_{\text{old}}$ .
11        Extend branches using  $S_g + \log \pi_{\text{old}} - \lambda_{\text{div}} d_g$  (Eq. 7).
12        Retain the top  $b$  extensions; reorder prefixes, caches, and ancestry.
13      End for; end for
14      Reorder prior trajectory records and memories by the selected ancestry.
15      Append each branch's frame; retain anchors and its recent history within  $B$ .
16    End for; return the highest-search-score trajectory from each group.
17    Compare: decode the  $G$  candidates and compute  $R_i$  (Eq. 8).
18    Compute  $A_i = (R_i - \bar{R}) / (\text{Std}(R) + \varepsilon)$ ; if  $\text{Std}(R) \leq 10^{-6}$ , set  $A_i = 0$ .
19    Learn: score candidate tokens under their saved generation-time histories.
20    Use these same conditions for old, current, and frozen reference policies.
21    Form  $\mathcal{J}_{\text{MemSPO}}$  (Eq. 11); take one optimizer step on  $-\mathcal{J}$ .
22  End for; return  $\theta$ .
```

Search penalties select candidates; unmodified policy probabilities score them.

Implementation correspondence. Search scores accumulate across the full horizon. Candidate selection also reorders previously saved frame prefixes and responses, preserving each final trajectory's ancestry. Reward computation and candidate construction are detached from differentiation; ground-truth futures enter rewards only. The reference penalty uses the numerically clipped form $\text{clip}(q - \log q - 1, -10, 10)$. This training search differs from evaluation-time Beam4, which selects one frame prediction before proceeding to the next frame.

B.3 BASELINE ADAPTATION AND EVALUATION

Shared evaluation. Table 1 evaluates 32-frame predictions on 128 matched trajectories per dataset. RT-1 and BridgeV2 use held-out development sets; RoboCasa uses a held-out test set. Predictions are resized to the reference resolution, and LPIPS-VGG, SSIM, and frame-wise PSNR are averaged over frames and trajectories. Recorded actions condition generation; ground-truth future images are evaluation targets and never refresh rollout histories. Models retain their native input interfaces and decoding procedures. Cosmos-Predict2.5-2B uses one initial observation, whereas FutureWorlds uses two. Ten-frame baseline results use prefixes of the same 32-frame rollouts.

FutureWorlds checkpoints. Main-table evaluations use a beam width of four and persistent memory, with 500/200/400 MemSPO updates for RT-1/BridgeV2/RoboCasa. BridgeV2 uses the R1 reward and omits text conditioning. The matched post-training comparison instead uses 200 updates for both RL methods on every dataset.

iVideoGPT and PersistWorld. On BridgeV2, iVideoGPT adapts the official OXE-256 tokenizer and autoregressive backbone for 20,000 updates each, freezing the tokenizer during backbone training. It uses two observations, recorded seven-dimensional actions, temperature 1, and top- k 100 sampling. Ten-frame chunks condition subsequent chunks on the two latest generated frames. PersistWorld starts from the released Ctrl-World checkpoint, performs 2,000 target-domain updates followed by 30 NFT updates, and uses generated latent history with 25 denoising steps and guidance scale 2.0. Its visual tokenizer and CLIP encoders remain frozen during base adaptation. Neither rollout accesses future ground-truth frames.

Wan2.2 adaptation. We adapt the official Wan2.2-Ti2V-5B checkpoint (Team Wan et al., 2025) using action and history adapters plus rank-32 LoRA (Hu et al., 2021) in attention layers (Figure 8). Following chunk-wise conditioning (Gao et al., 2026), four actions modulate each future latent step; the previous observation supplies spatial history. The latest observation remains a clean anchor. Only adapters and LoRA parameters are trained with future-latent flow matching; the pretrained transformer, VAE, and text encoder remain frozen. The BridgeV2 run uses two observed and twelve future frames, training-only action normalization, and AdamW with a cosine learning-rate schedule from 10^{-4} to 10^{-5} . A disjoint 32-example monitor selects the 8,000-update checkpoint using LPIPS-Alex; the final 128-example evaluation uses LPIPS-VGG. Inference uses 50 UniPC steps (Zhao et al., 2023), with guidance scale and time shift both set to 5.

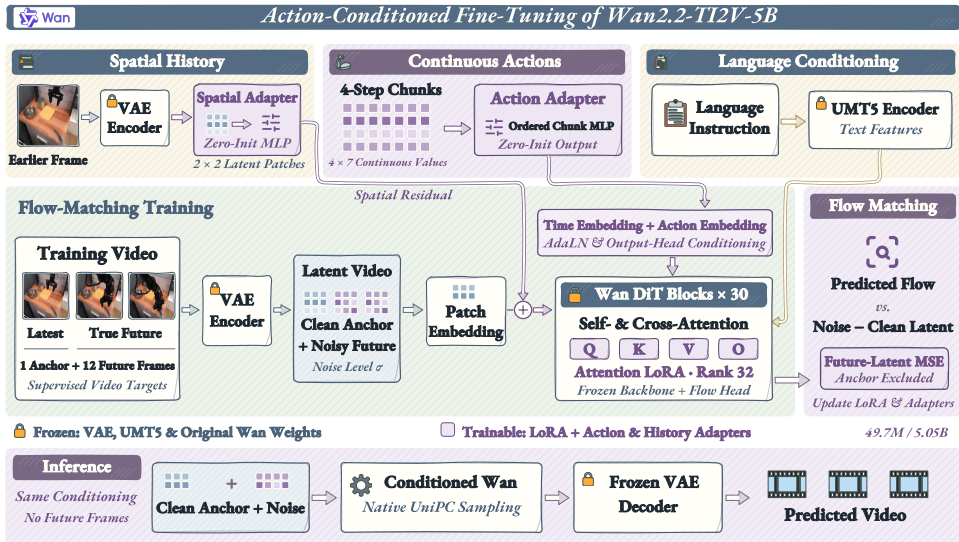


Figure 8: **Action-conditioned Wan2.2 adaptation.** Action chunks modulate time embeddings, and a history adapter supplies spatial context. Adapters and LoRA parameters are trainable; pretrained weights remain frozen.

Release plan. We plan to release all implementation code and trained checkpoints, including adaptation configurations and evaluation scripts.

B.4 LEARNING-CURVE AND QUALITATIVE EVALUATION

Figure 4(a,b) evaluates 32 fixed monitoring clips per dataset with 10-frame prediction and a beam width of four. All checkpoints share inputs and decoding settings. Circles show measurements, curves are endpoint-preserving smoothed guides, and stars mark 200-update MemSPO checkpoints. SFT and RL use different horizontal scales; update counts do not imply equal compute.

Figure 4(c) shows selected improved examples at frame 32. The RT-1, BridgeV2, and RoboCasa models use 500, 200, and 400 MemSPO updates, respectively; the BridgeV2 example uses the R0 checkpoint underlying the learning curve. Each SFT comparison matches inputs and decoding. Full frames remain visible, and detail crops use identical coordinates across methods. These examples illustrate prediction quality rather than measured robot task success.

B.5 DECODING SENSITIVITY

Table 3: LPIPS reduction (%) across decoding settings.

Temperature T	Top- p	RT-1	BridgeV2
1.0	1.00	4.02	3.45
0.8	1.00	3.03	2.97
1.2	1.00	4.32	2.60
1.0	0.95	3.70	3.84
1.0	0.90	4.04	3.53

Decoding sensitivity. Table 3 varies temperature T and the nucleus threshold p , keeping the beam width at four. On the same 32 clips per dataset, MemSPO reduces mean LPIPS under all five settings: 3.03–4.32% on RT-1 and 2.60–3.84% on BridgeV2. Each setting uses its own matched SFT reference, so the gains do not depend solely on the default decoding parameters.

B.6 MOTION AND DYNAMIC EVALUATION

Figure 6 uses the same 128 held-out trajectories per dataset and scores 32 predicted frames at 256×320 resolution. We use WorldArena’s RAFT and DINO components (Shang et al., 2026) to construct three reference-based diagnostics. Flow is the mean Euclidean difference between predicted-video and ground-truth optical-flow vectors over adjacent frames and spatial positions. Gap is the absolute difference between their mean flow magnitudes. DINO is the mean cosine similarity between corresponding predicted and ground-truth frame features. Each metric is averaged over trajectories; lower Flow and Gap and higher DINO indicate closer agreement with the recorded future. These diagnostics are supplementary measures rather than the official WorldArena aggregate score.

All metrics use GPU FP32 computation with TF32 disabled. FutureWorlds uses the main-comparison checkpoints: 500, 200 (R1), and 400 MemSPO updates for RT-1, BridgeV2, and RoboCasa. RT-1 and RoboCasa videos are regenerated with the original decoding settings, while BridgeV2 retains the saved prediction tokens and decodes them uniformly on GPU. This supplementary evaluation does not replace the historical pixel-metric results. Ten- and twenty-frame evaluations use prefixes of the same rollouts. The qualitative cases are selected improved examples. For each case, the visualized pair has the largest ground-truth mean motion among the three stored pairs beginning at frames 8, 16, and 31. All methods share that pair and a color scale set by the pooled 99th-percentile flow magnitude; larger magnitudes are clipped. Hue represents direction and saturation represents magnitude. Values in the main figure are rounded to two decimal places.

C SHORT-HORIZON EVALUATION

Table 4 complements the 32-frame comparison with the available ten-frame results, using the same model checkpoints and 128 trajectories per reported dataset. Baseline scores use the first ten frames of their 32-frame rollouts; FutureWorlds uses its existing ten-frame evaluations. We average frame-wise PSNR, SSIM, and LPIPS-VGG over each trajectory and then over trajectories.

Table 4: Ten-frame video prediction on 128 trajectories per reported dataset. Metrics: PSNR $\uparrow$, SSIM $\uparrow$, and LPIPS $\downarrow$. Dashes indicate unavailable results.

Method	RT-1			BridgeV2			RoboCasa		
	PSNR	SSIM	LPIPS	PSNR	SSIM	LPIPS	PSNR	SSIM	LPIPS
<i>Video Generation Models</i>									
FitVid	–	–	–	17.86	0.6241	0.5079	14.29	0.5640	0.5373
iVideoGPT	–	–	–	21.91	0.8142	0.1544	17.56	0.7513	0.2009
<i>Action-Conditioned World Models</i>									
IRASim	–	–	–	18.65	0.7442	0.2120	14.75	0.6495	0.3069
Vid2World	–	–	–	–	–	–	–	–	–
Ctrl-World	20.67	0.7548	0.2094	20.84	0.7456	0.2071	16.27	0.7335	0.2325
DriftWorld	–	–	–	22.72	0.8103	0.1699	–	–	–
WEAVER	21.12	0.8038	0.1932	21.20	0.7929	0.1993	15.84	0.7523	0.2308
<i>RL Post-Training</i>									
RLVR-World	–	–	–	–	–	–	–	–	–
PersistWorld	–	–	–	21.62	0.7646	0.1907	16.97	0.7481	0.2166
<i>Adapted Video Foundation Models</i>									
Wan2.2-TI2V-5B	20.12	0.7707	0.1892	19.03	0.7784	0.1737	–	–	–
Cosmos-Predict2.5-2B	–	–	–	21.36	0.8365	0.1301	15.51	0.7382	0.2268
DreamDojo-2B	–	–	–	19.31	0.8190	0.1808	15.10	0.7454	0.2322
FutureWorlds (ours)	24.21	0.8345	0.1252	24.34	0.8318	0.1072	20.18	0.8045	0.1434

Native interfaces. IRASim on BridgeV2 receives recorded commands, whereas its released checkpoint is trained with state-derived relative motion. Its RoboCasa result instead uses a target-domain 52-dimensional action adapter. Missing ten-frame evaluations remain unreported rather than being inferred from longer-horizon scores.

D POLICY FEEDBACK IN A LEARNED WORLD MODEL

We examine a selected RT-1 drawer-closing example with a frozen, converged RT-1 policy. Starting from the same two recorded observations, the policy acts on each model’s latest generated frame, and the world model predicts the next observation from that action. SFT, GRPO, and MemSPO share the initial policy state, random seed, beam width of four, and memory configuration; both post-trained models use 200 updates. No recorded future frame refreshes the interaction.

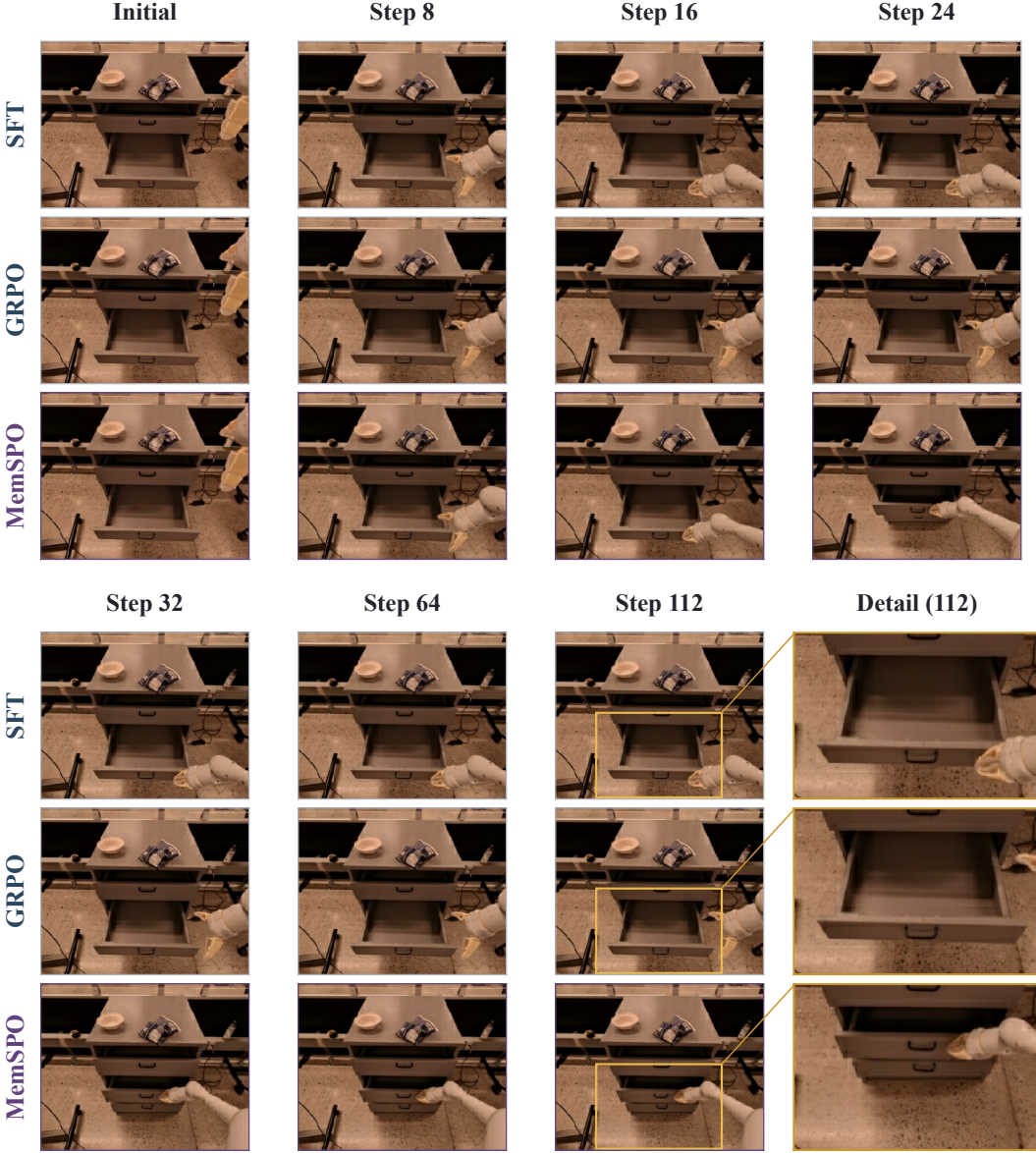


Figure 9: **Policy feedback in a learned world model.** A selected example for *close middle drawer* with a fixed RT-1 policy. The two blocks show the same interaction at seven time points; gold boxes mark matched details at step 112.

Figure 9 shows a visibly smaller drawer opening under MemSPO from step 32 onward, while SFT and GRPO retain larger openings. The contrast illustrates how post-training changes the scene evolution observed by the same policy during feedback interaction. This qualitative example concerns learned-model behavior and does not establish task success in a physical environment.